178066: recrystallized sandstone, Minga Bore

Location and sampling

YOUANMI (SH 50-4), EVERETT CREEK (2841)
MGA Zone 50, 775270E 6849310N

Sampled on 31 October 2002

The sample was taken from a 0.5 m-diameter block about halfway up the northwestern side of a northwest-trending rocky ridge, 100 m southeast of the access track and 0.5 km west of Minga Bore. The sampling site is 50 m east of the site of GSWA 178065 (dated by Nelson, this compilation).

Tectonic unit/relations

The sample is a massive, greyish-white, coarse- and even-grained, recrystallized quartz sandstone from a bedded quartzite unit in the Maynard Hills greenstone belt of the Southern Cross Granite–Greenstone Terrane, Yilgarn Craton. There is no established stratigraphy for this belt (Riganti, 2003). Bedding in the quartzite unit strikes at 340° and dips steeply to the southwest. The sample was taken 50 m stratigraphically below the unit from which sample 178065 was collected, and contains fractures, up to 1 mm thick, possibly containing quartz veins. The sample was collected to constrain the maximum depositional age for the sandstone precursor to the quartzite.

Petrographic description

The principal mineral is quartz (99 vol.%), with minor muscovite (<1 vol.%), and trace amounts of accessory rutile and zircon. The sample is a massive quartzite consisting of anhedral quartz grains up to 7 mm long, with fluid inclusions and some stylolitic grain-boundaries. Minor mica up to 0.15 mm in grain size is weakly aligned, but quartz c-axes mostly fall within a 90° arc across the thin section. The metamorphic grade and protolith are uncertain.

Zircon morphology

The zircons isolated from this sample are typically colourless, pale yellow or dark brown, elliptical or subhedral and with rounded terminations, equant to slightly elongate, and between 50 × 80 µm and 160 × 80 µm in size. Most grains are fractured and internally structureless, but some have faint internal zonation. Many have a speckled appearance in transmitted light, and many are metamict. Many grain surfaces are pitted, consistent with detrital transport. In reflected light, many surfaces have irregular, dark patches and zones. Cathodoluminescence images of representative zircons are given in Figure 1.

Analytical details

This sample was analysed on 1 and 17 April 2003. The counter deadtime during both analysis sessions was 24 ns. During the first analysis sesssion, three analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 2.06% (1σ). Analyses 1.1 to 7.1 were obtained during the first analysis session. During the second analysis session, five analyses of the CZ3 standard indicated a Pb*/U calibration uncertainty of 1.74% (1σ). Common-Pb corrections were applied assuming Broken Hill common-Pb isotopic compositions for all analyses, with the exception of analyses 26.1 and 30.1, for which isotopic compositions determined using the method of Cumming and Richards (1975) were assumed.

Results

Thirty-five analyses were obtained from 35 zircons. Results are given in Table 1, and shown on concordia and Gaussian-summation probability density plots in Figures 2 and 3 respectively.

Interpretation

The analyses are concordant to highly discordant, with the discordance pattern consistent with a single recent episode of radiogenic-Pb loss. On the basis of their $^{207}Pb/^{206}Pb$ ratios, many analyses can be assigned to one of three groups. Twenty-five concordant to highly discordant analyses of 25 zircons (1.1, 2.1, 3.1, 4.1, 5.1, 6.1, 7.1, 8.1, 11.1, 13.1, 14.1, 15.1, 16.1, 17.1, 18.1, 20.1, 21.1, 22.1, 23.1, 24.1, 25.1, 26.1, 27.1, 29.1, 31.1, 35.1), assigned to Group 1, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3374 ± 3 Ma (chi-squared = 1.43). Concordant analyses 19.1 and 28.1, assigned to Group 2, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3269 ± 2 Ma (± 1σ uncertainty). Concordant analyses 10.1 and 12.1, assigned to Group 3, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean $^{207}Pb/^{206}Pb$ date of 3331 ± 4 Ma (± 1σ uncertainty). The remaining analyses cannot be confidently grouped.

The analyses are interpreted to be of detrital zircons. The date of 3269 ± 2 Ma (± 1σ uncertainty) indicated by the weighted mean $^{207}Pb/^{206}Pb$ ratio of concordant analyses 19.1 and 28.1 of Group 2 is therefore interpreted as providing a maximum age for deposition of the precursor sandstone. Source rocks within the western part of Australia having ages matching those of the zircons within this sample include the Narryer Terrane in the northwestern part of the Yilgarn Craton (Nelson, 1995, 1996, 1997) and the eastern part of the Pilbara Craton (Nelson, 1996, 1997, 1998, 1999, 2000, 2001, 2002, this compilation).

Recommended reference for this publication:

NELSON, D. R., 2005, 178066: recrystallized sandstone, Minga Bore; Geochronology dataset 593; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

Data obtained: 17/04/2003; Data released: 30/06/2005

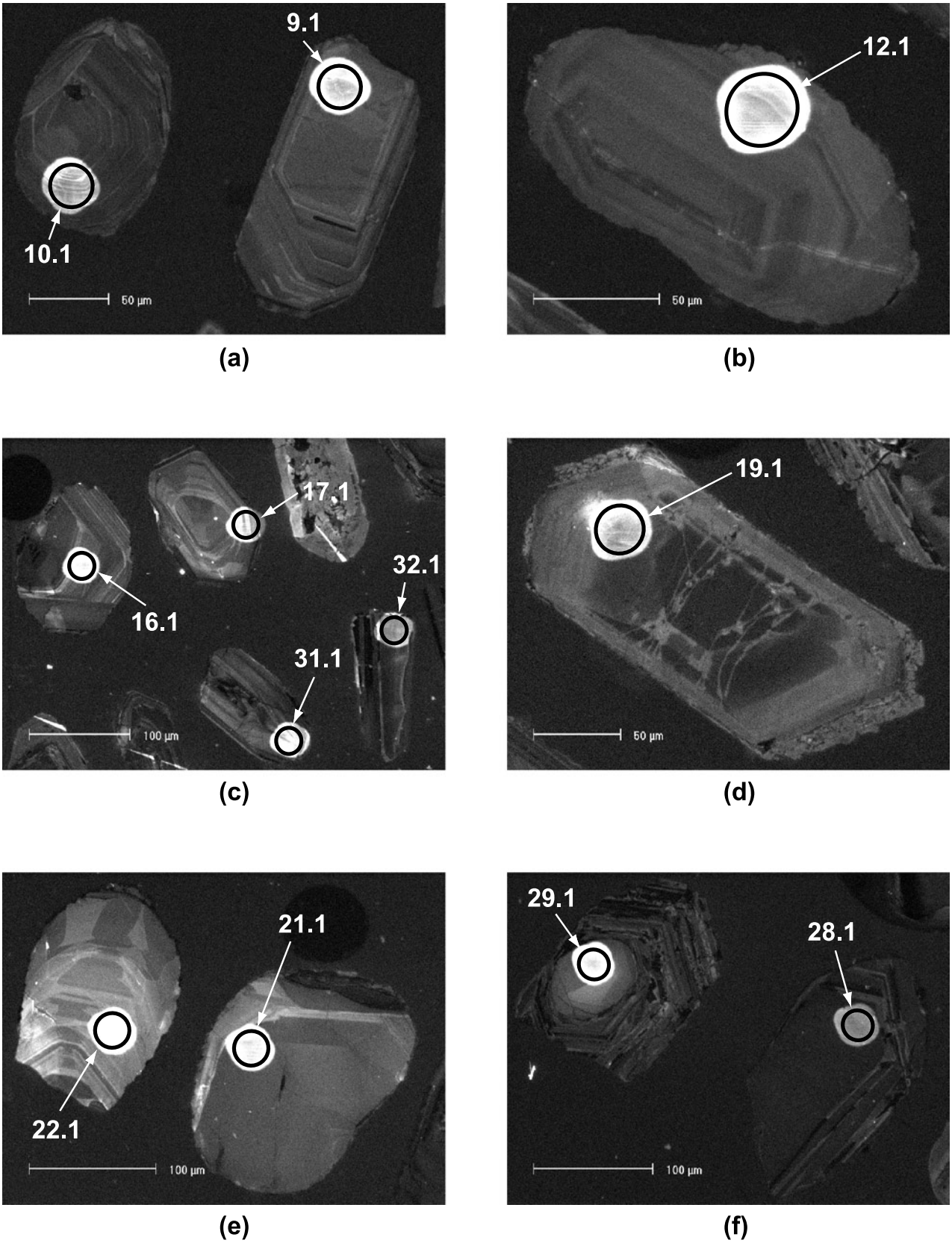

Figure 1. Cathodoluminescence images of representative zircons from sample 178066: recrystallized sandstone, Minga Bore

Table 1. Ion microprobe analytical results for zircons from sample 178066: recrystallized sandstone, Minga Bore

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	±1σ	$^{208}Pb/^{206}Pb$	±1σ	$^{206}Pb/^{238}U$	±1σ	$^{207}Pb/^{235}U$	±1σ	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	±1σ
1.1	178	133	148	0.082	0.28096	0.00114	0.19805	0.00135	0.6540	0.0138	25.335	0.556	96	3 368	6
2.1	245	161	205	0.234	0.27997	0.00107	0.17962	0.00137	0.6617	0.0138	25.544	0.555	97	3 363	6
3.1	246	162	195	0.132	0.28290	0.00101	0.17324	0.00117	0.6310	0.0132	24.611	0.534	93	3 379	6
4.1	176	52	137	0.253	0.29715	0.00126	0.07634	0.00129	0.6519	0.0137	26.710	0.588	94	3 455	7
5.1	127	41	97	0.209	0.28215	0.00135	0.08254	0.00138	0.6454	0.0137	25.109	0.559	95	3 375	7
6.1	183	192	171	0.125	0.28405	0.00108	0.27552	0.00150	0.6941	0.0146	27.186	0.595	100	3 385	6
7.1	76	34	64	0.366	0.28255	0.00177	0.12466	0.00232	0.6882	0.0149	26.810	0.623	100	3 377	10
8.1	69	109	70	0.106	0.28436	0.00135	0.40909	0.00227	0.6914	0.0238	27.110	0.958	100	3 387	7
9.1	117	61	101	0.067	0.27877	0.00103	0.13650	0.00108	0.7058	0.0242	27.130	0.947	103	3 356	6
10.1	187	109	154	0.055	0.27449	0.00082	0.15356	0.00088	0.6693	0.0229	25.332	0.878	99	3 332	5
11.1	132	39	110	0.047	0.28135	0.00092	0.07765	0.00071	0.7139	0.0244	27.695	0.963	103	3 370	5
12.1	100	101	93	0.115	0.27405	0.00111	0.27106	0.00159	0.6982	0.0239	26.381	0.924	103	3 329	6
13.1	220	87	183	0.061	0.28162	0.00074	0.10275	0.00067	0.6987	0.0238	27.130	0.937	101	3 372	4
14.1	92	77	66	0.128	0.28227	0.00127	0.14257	0.00147	0.5812	0.0199	22.621	0.795	88	3 376	7
15.1	199	119	168	0.027	0.28250	0.00075	0.14784	0.00072	0.6853	0.0234	26.695	0.923	100	3 377	4
16.1	87	42	75	0.053	0.28254	0.00117	0.12885	0.00117	0.7060	0.0242	27.503	0.965	102	3 377	6
17.1	128	113	116	0.207	0.28101	0.00101	0.23330	0.00144	0.6943	0.0237	26.900	0.937	101	3 369	6
18.1	61	28	52	0.178	0.28210	0.00147	0.11839	0.00166	0.6964	0.0240	27.087	0.962	101	3 375	8
19.1	931	1 293	846	0.025	0.26346	0.00036	0.36916	0.00058	0.6453	0.0220	23.440	0.802	98	3 268	2
20.1	141	56	117	0.119	0.28004	0.00097	0.10461	0.00103	0.6935	0.0237	26.777	0.932	101	3 363	5
21.1	125	70	105	0.031	0.28201	0.00098	0.14778	0.00094	0.6836	0.0234	26.582	0.926	100	3 374	5
22.1	33	47	32	0.521	0.27949	0.00229	0.37975	0.00413	0.6763	0.0236	26.064	0.961	99	3 360	13
23.1	86	55	73	0.075	0.28222	0.00122	0.17100	0.00134	0.6835	0.0235	26.596	0.935	99	3 375	7
24.1	129	58	107	0.113	0.28088	0.00101	0.11810	0.00107	0.6845	0.0234	26.511	0.924	100	3 368	6
25.1	158	130	91	0.230	0.28227	0.00114	0.28161	0.00176	0.4236	0.0145	16.486	0.575	67	3 376	6
26.1	132	84	92	0.883	0.27858	0.00138	0.20196	0.00232	0.5287	0.0181	20.308	0.714	82	3 355	8
27.1	110	38	90	0.054	0.28440	0.00108	0.09305	0.00091	0.6847	0.0235	26.849	0.938	99	3 387	6
28.1	301	69	226	0.016	0.26433	0.00064	0.06230	0.00044	0.6572	0.0224	23.951	0.826	99	3 273	4
29.1	78	46	68	0.187	0.28435	0.00136	0.15821	0.00168	0.6994	0.0241	27.421	0.968	101	3 387	7
30.1	939	1 186	247	0.732	0.25028	0.00075	0.17774	0.00124	0.2089	0.0071	7.210	0.249	38	3 187	5
31.1	111	101	100	0.141	0.28118	0.00111	0.24374	0.00148	0.6823	0.0234	26.453	0.925	100	3 370	6
32.1	311	169	259	0.064	0.28438	0.00063	0.14417	0.00064	0.6765	0.0231	26.526	0.913	98	3 387	3
33.1	220	181	192	0.051	0.27216	0.00074	0.21787	0.00090	0.6772	0.0231	25.411	0.878	100	3 319	4
34.1	478	384	269	0.207	0.28918	0.00063	0.14962	0.00074	0.4523	0.0154	18.033	0.620	70	3 413	3
35.1	109	98	99	0.073	0.28204	0.00109	0.23960	0.00139	0.6874	0.0236	26.732	0.935	100	3 374	6

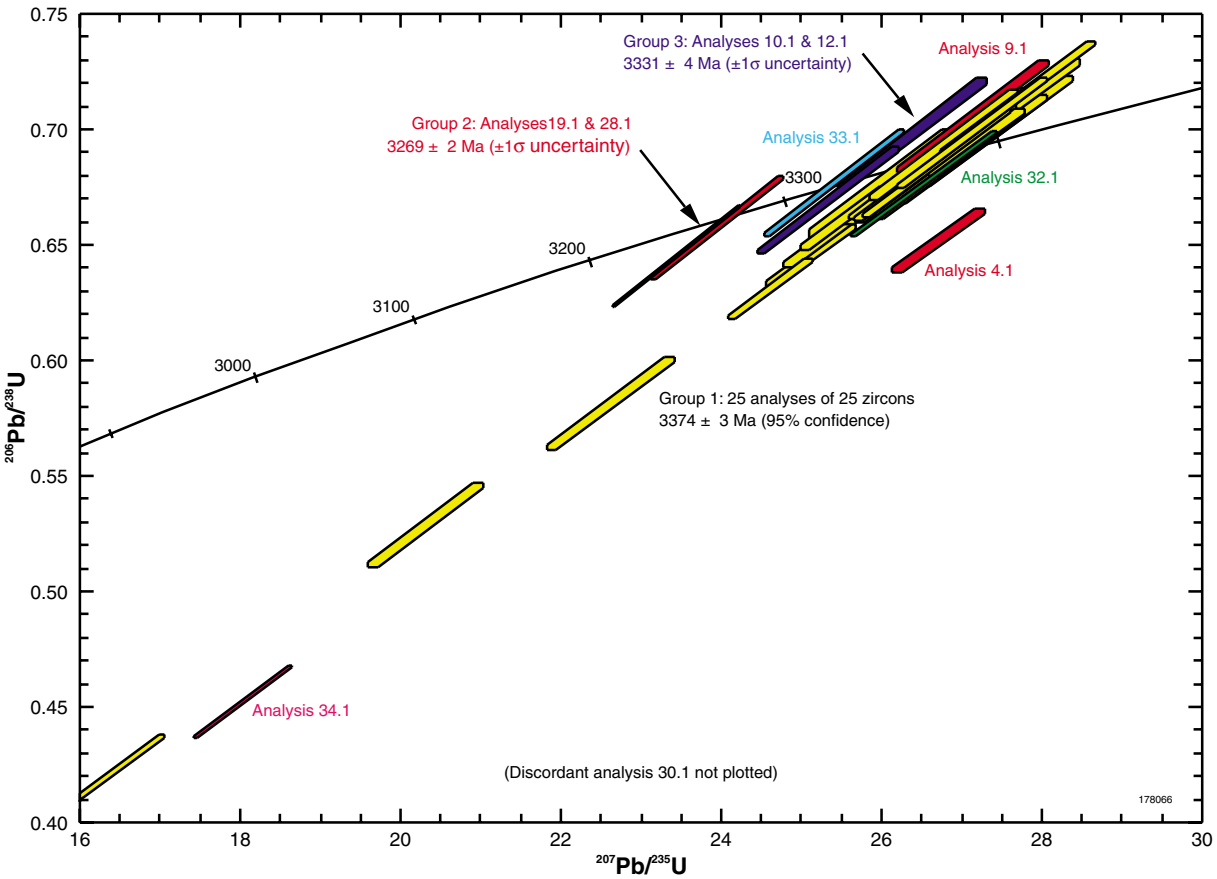

Figure 2. Concordia plot for sample 178066: recrystallized sandstone, Minga Bore

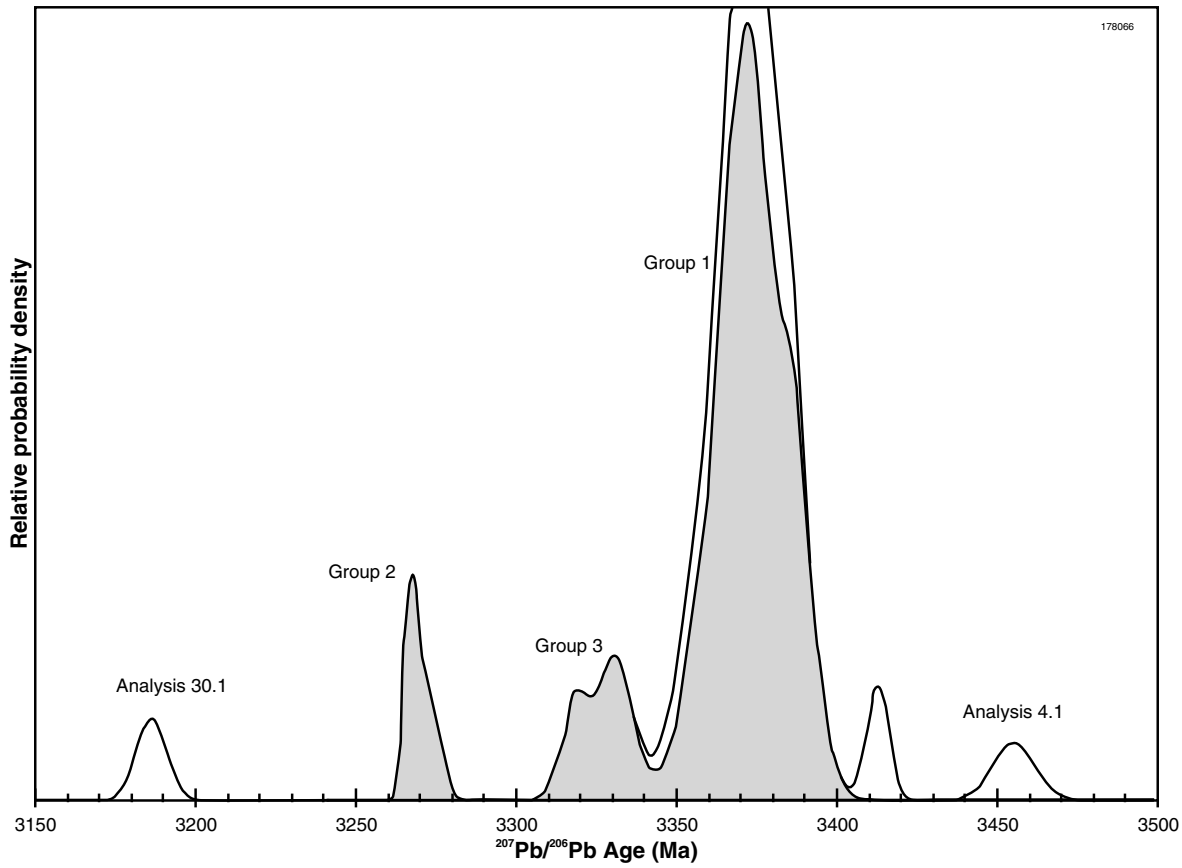

Figure 3. Gaussian-summation probability density plot for sample 178066: recrystallized sandstone, Minga Bore